Yale JACKSON SCHOOL OF GLOBAL AFFAIRS

Final Data Project Due Monday, May 4th

GLBL 5021 Applied Methods of Analysis II

Students conduct a full data analysis project: (i) develop a question, (ii) find the appropriate data, (iii) choose a methodology, (iv) create visualizations and tables, and (v) write-up key results. Along the way, you will likely have to troubleshoot many issues or tweak your data analysis plan. But you will learn a lot throughout the process and have an additional output as part of your portfolio.

Group vs Individual Report

The final project can be done individually or in groups of two (max). Those in groups of two need to submit a write-up on task division and make sure they complete twice the data analysis requirements (will need to have two different analyses and double visualization/tables).

Finding Data

First, you want to decide whether to do a macro-level question or a micro-level question and what would your unit of analysis be (individual, household, country, firm, hotel vacancy on a particular date, etc.). Then, a good way to check what data is available is to find papers discussing the topic and checking the data they are using. What type of questions are people interested in? What questions haven't been explored? What type of variables are important for the topic or question of interest? Brainstorm with a causal map to help you narrow down variables (even if you will do a predictive methodology later, this is a good start). If you are confident, you can try web-scraping your data with the help of Claude. If you are having trouble narrowing down a dataset, please email the instructor for advice.

Here are examples of datasets. You can use multiple data sources.

i) Macro-level Dataset

- Penn World Tables
- World Development Indicators Data Bank (for a list of indicators) or access through this link
- You could also combine the above indicators with more specific indicators depending on your topic. Topic specific data:
    - Worldwide Governance Indicators
    - AidData
    - G^2LM|LIC - Jobs of the World Project
    - Trade Data

ii) Micro-level Dataset (it is better to choose one country of interest)

- Household surveys that can be accessed from the World Bank microdata
    - Living Standard Measurement Survey
    - High Frequency Phone Survey

    - Enterprise Survey (this is data on firms)
- Datasets from IPUMS

Format and Components of the Final Write Up

A good write-up is concise, transparent, and effective in presenting findings to its intended audience. A brief report that packs a lot of information concisely is always better than a rambling report. Be concise and kind to your audience.

The report for this class will have several parts.

Part I. Title and Executive Summary

- This section consists of just one paragraph or several bullet points of a summary and key findings of the write-up.

Part II. Question Motivation & Background

- This section consists of three paragraphs (max) that describe the policy, economic or business question of interest and background motivation. Why is this question important to answer? For each paragraph, make sure you have a key sentence that is highlighted in the beginning of the paragraph.

Part III. Data Description

- This section consists of three paragraphs (max) that describe the dataset you are using and key variables.
- You need to include a minimum of one: table of descriptive statistics of key variables or one graph showing patterns of key variables. If more appropriate, you can move the graph from the data description section to the motivation/background section.

Part III. Methodology

- This section consists of three paragraphs (max) that describe what type of methodology or analysis you will use to answer your question of interest. Do you care about association or prediction? What is the model that you will use (describe the equation)? Why does this choice of methodology work?

Part IV. Key Results

- You can break this up into multiple sections depending on the different types of analysis that you have. For example, Part 4a compares models for predicting hotel prices and Part 4b discusses the result list on which hotels are cheaper and how consumers should choose their hotels. If you are in a group of two, two types of analysis are necessary.
- You should have a minimum of one table AND a minimum of one graph in the key result section.

Part V. Summary Conclusion

- Summarize your analysis in one or two paragraphs. You can note further questions to explore stemming from your analysis and practical use of analysis in the future.

Part VI. References

This is a list of references used for the report. Should have at least a few references especially when discussing the background question and/or in the policy implication discussion.

Some notes about formatting:

- You can use any software for your write-up. A simple word document is fine. It is good practice, however, to format your final write-up to look professional and aesthetically appealing using Canva or Microsoft Publisher. It doesn't have to be fancy, but a clean design is preferred.

Grading Criteria

The guidelines below are ONLY the grading criteria for the final submission of the report. Throughout the course, you do have other components that are part of the grade such as the draft proposal, draft graphs, and other tasks related to the project. But for the final submission itself, I will grade based on:

(1) completeness (having all sections and required components)
(2) professionalism (how does the report looks; formatting; graphs and tables labeled properly)
(3) methodology choice and logic of analysis (chose an appropriate methodology for the question, correctness in applying the methodology, appropriate choice of graphs and tables that flows well)
(4) writing effectiveness (clarity and conciseness of writing)